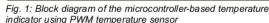


Different methods for interfacing the sensors with processing units have evolved over the years. Each of them is aimed at reducing the input/output (I/O) interface lines between the sensor and the processing unit without sacrificing the resolution and accuracy of measurement.

is one such method of interfacing. In this method, the information regarding the measured quantity of data is passed to the processing unit by means of a varying pulse width (or duty cycle or pulse frequency). It uses only one I/O interface line with the microcontroller. A number of sensors employing this technique of interfacing are available for measuring such parameters as temperature, gravity and voltage.

crocontrollers are equipped with internal timers and a variety of features such as prescaler (which increases the range of measurement) and capture. Internal timer units can be used to measure



pulse width, pulse frequency and duty cycle, thus simplifying the circuit for measurement.

Fig. 1 shows the block diagram of the microcontroller-based temperature indicator using PWM temperature sensor. A very simple temperature monitor/indicator can be built around temperature sensor MAX6577 and 8-bit RISC microcontroller ATmega16. Temperature sensor MAX6577 converts the temperature into a square wave and is capable of measuring temperature in the range of -40°C to $+125^{\circ}\text{C}$ without requiring any external component/sub-system for the purpose. Microcontroller ATmega16 has all the features

for allowing direct interfacing of the temperature sensor. It measures the frequency of the signal received from the temperature sensor, processes it and displays the temperature in °C on a liquid-crystal display (LCD) module.

Circuit description

Fig. 2 shows the circuit of the microcontroller-based temperature indicator using PWM temperature sensor. At the heart of